From Critical Coherence to Structural Selection: A Unified Order-Parameter and Functional Framework

Christof Krieg

Independent Researcher

maat-research.com

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Abstract

We propose and formalise a two-level structural description of dynamical systems built on five diagnostic sectors $\{H, B, S, V, R\}$: equation-solution consistency, energy-momentum balance, dynamic activity, inter-field connectivity, and geometric robustness. The *Critical Coherence Index* (CCI) condenses these sectors into a single ratio that acts as an order parameter distinguishing ordered, critical, and chaotic dynamical regimes. The *MAAT structural functional* F_{MAAT} is a log-sum penalty that preserves sector-wise information and acts as a structural analogue of a free-energy functional. We show that CCI is a coarse-grained projection of the same structural state vector underlying F_{MAAT}, formalise their relationship within a Landau-type framework, and derive explicit conditions under which minimisers of F_{MAAT} are contained within a prescribed CCI regime. Empirical interpretations are given for nonlinear field systems (soliton ranking, chaos onset) and cosmological toy models (expansion history selection). The framework provides a principled hierarchy: dynamics determines admissible states, CCI classifies them by regime, and F_{MAAT} selects structurally preferred configurations within each regime.

Contents

1 Introduction **2**

2 Structural Sectors and Sector Maps **2**
2.1 The five structural sectors 2

3 The MAAT Structural Functional **3**

4 CCI as an Order Parameter **4**

5 Relationship Between CCI and F_{MAAT} **5**
5.1 CCI as a projection 5
5.2 Landau-type expansion 5

6 Structural Selection Results **6**

7 Empirical Interpretation **6**
7.1 Nonlinear field systems 6
7.2 Cosmological expansion histories 7

8 Relation to Existing Variational Principles **7**

9 Discussion **8**

10 Conclusion **9**

A Worked Example: Two-Sector Reduction **10**

B Derivation of the Landau Coefficients **10**

1 Introduction

A central challenge in the study of complex dynamical systems is to characterise not only *which* states are physically admissible, but *which among them are structurally preferred*. Traditional tools—Lyapunov exponents, correlation lengths, bifurcation diagrams—classify dynamical regimes but do not provide a continuous ranking of structural quality across configurations within the same regime.

Two complementary diagnostics have been introduced in the context of multi-field coherence theory (1; 2) to address this gap:

(i) the *Critical Coherence Index* (CCI), a dimensionless ratio that distinguishes ordered, critical, and chaotic behaviour;

(ii) the *MAAT structural functional* F_{MAAT}, a log-sum penalty over the same structural sectors that provides a fine-grained ranking of configurations.

Although both are derived from the same five structural sectors, their mathematical roles differ: CCI performs a projection onto a one-dimensional order parameter, while F_{MAAT} retains the full sector-wise information. This paper establishes their precise relationship, situates it within a Landau-type theory, and derives its consequences for structural selection.

The paper is organised as follows. Section 2 defines the structural sectors and their bounded representatives. Section 3 introduces F_{MAAT} as a structural free energy. Section 4 introduces CCI as an order parameter and characterises the induced regime partition. Section 5 shows that CCI and F_{MAAT} are distinct projections of the same structural state vector and derives the Landau-type expansion. Section 6 states the main structural selection results. Section 7 discusses empirical interpretations. Section 8 situates the framework among existing variational principles. Section 10 concludes.

2 Structural Sectors and Sector Maps

2.1 The five structural sectors

Let $\mathbf{\Phi} = (\Phi_1, \ldots, \Phi_n)$ be a collection of dynamical fields on a spacetime manifold $\mathcal{M}$. We associate to any field configuration five non-negative *defect measures*:

$$d_H = \sum_i \left(\frac{\delta \mathcal{S}}{\delta \Phi_i}\right)^2, \tag{1}$$

$$d_B = (\nabla_\mu T^{\mu\nu})(\nabla_\rho T^\rho{}_\nu), \tag{2}$$

$$d_S = \left[\frac{\Lambda_C^4}{\eta_C(\Box\Phi)^2 + \Lambda_C^4}\right], \tag{3}$$

$$d_V = \left[\sum_{i<j} \xi_{ij}\, \nabla_\mu \Phi_i\, \nabla^\mu \Phi_j\right]^{-1}, \tag{4}$$

$$d_R = \alpha_1(\nabla_\mu G^{\mu\nu})^2 + \alpha_2(\nabla_\mu T^{\mu\nu})^2. \tag{5}$$

Here d_H measures deviation from on-shell equations of motion; d_B measures violation of energy-momentum conservation; d_S is an *inverse activity defect*: the formula (3) is large (close to 1) when $(\Box\Phi)^2 \ll \Lambda_C^4$ (low activity) and small when $(\Box\Phi)^2 \gg \Lambda_C^4$ (high activity), so high dynamic activity corresponds to small d_S and hence large Γ_S; d_V penalises weak inter-field coupling; and d_R measures geometric-material consistency violations. The constants $\Lambda_C, \eta_C, \xi_{ij}, \alpha_{1,2}$ are model parameters.

Definition 1 (Sector maps)**.** For each sector $a \in \{H, B, S, V, R\}$, the *sector map* Γ_a is defined by

$$\Gamma_a = \frac{1}{1 + d_a} \;\in\; (0,\, 1]. \tag{6}$$

$\Gamma_a = 1$ corresponds to a structurally perfect configuration in sector a; $\Gamma_a \to 0$ corresponds to a maximally defective one.

The five-tuple $\mathbf{\Gamma} = (\Gamma_H, \Gamma_B, \Gamma_S, \Gamma_V, \Gamma_R) \in (0, 1]^5$ constitutes the *structural state vector* of a field configuration.

Remark 1 (Operationalisation)**.** The general defect measures (1)–(5) are continuum expressions valid for smooth field configurations. In concrete numerical models they are replaced by operationalised proxys appropriate to the discretisation: for example, Friedmann equation residuals in FLRW toy models, discrete Laplace operators in lattice field theory, or finite-difference approximations to energy-momentum divergences. The sector-map formula (6) applies unchanged; only the inputs d_a differ.

3 The MAAT Structural Functional

Definition 2 (MAAT structural functional)**.** Let $\boldsymbol{\lambda} = (\lambda_H, \lambda_B, \lambda_S, \lambda_V, \lambda_R)$ be positive weights and $\epsilon > 0$ a regularisation parameter chosen for numerical stability (typical values: $\epsilon = 10^{-8}$ in floating-point implementations; $\epsilon = 0.01$ in the worked examples of Appendix A for legibility). The *MAAT structural functional* is

$$F_{\mathrm{MAAT}}(\mathbf{\Gamma}) = -\sum_{a \in \{H,B,S,V,R\}} \lambda_a \log\left(\frac{\epsilon + \Gamma_a}{\epsilon + 1}\right). \tag{7}$$

Remark 2. The normalisation by $(\epsilon + 1)$ ensures that the maximum $\Gamma_a = 1$ contributes zero to the sum, so $F^*_{\mathrm{MAAT}} = 0$ exactly at the structurally perfect state (all $d_a = 0$). Any structural defect ($\Gamma_a < 1$) raises the argument of the logarithm below unity, giving a positive contribution. As $\Gamma_a \to 0^+$, the contribution diverges, so structural defects in any sector are penalised without bound.

The functional has the following properties.

Proposition 1 (Properties of F_{MAAT})**.** *Let $\epsilon \in (0,1)$. Then:*

(i) ***Non-negativity.*** *$F_{\mathrm{MAAT}} \geq 0$ for all $\mathbf{\Gamma} \in (0,1]^5$, with $F_{\mathrm{MAAT}} = 0$ if and only if all $\Gamma_a = 1$.*

(ii) ***Additivity.*** *F_{MAAT} decomposes as a sum over independent sector contributions.*

(iii) ***Monotonicity.*** *F_{MAAT} is strictly decreasing in each Γ_a, so improving any single sector strictly lowers the functional.*

(iv) ***Convexity.*** *F_{MAAT} is strictly convex in $\mathbf{\Gamma}$ over $(0,1]^5$.*

Proof. (i) Since $\Gamma_a \in (0,1]$, we have $(\epsilon + \Gamma_a)/(\epsilon + 1) \in (0,1]$, so each logarithm is non-positive, and the negated sum is non-negative. Equality holds iff $\Gamma_a = 1$ for all a. (ii) is immediate from the definition. For (iii), $\partial F_{\mathrm{MAAT}}/\partial \Gamma_a = -\lambda_a/(\epsilon + \Gamma_a) < 0$. For (iv), the Hessian is diagonal with entries $\lambda_a/(\epsilon + \Gamma_a)^2 > 0$. □

We interpret F_{MAAT} as a *structural free energy*: just as a thermodynamic free energy singles out equilibrium states among all states consistent with macroscopic constraints, F_{MAAT} singles out structurally coherent configurations among all dynamically admissible ones.

4 CCI as an Order Parameter

Definition 3 (Critical Coherence Index — reduced sector form)**.** In this paper we employ a *reduced sector form* of the CCI, using the five MAAT sector maps directly:

$$\mathrm{CCI}(\mathbf{\Gamma}) = \frac{\Gamma_S}{\Gamma_H + \Gamma_B + \Gamma_V + \Gamma_R}, \tag{8}$$

i.e. the ratio of dynamic activity to the total stabilising structure. The canonical multi-field form used in earlier work (1; 2) employs additional weighting by instability and production factors; the present form is a minimal version sufficient to establish the two-level framework.

Remark 3 (Sector terminology)**.** The five sectors $\{H, B, S, V, R\}$ correspond across the paper series as follows:

Symbol	This paper	Earlier series
H	Equation consistency	Harmonic residual / equation balance
B	Energy-momentum balance	Conservation consistency
S	Dynamic activity	Structural activity / production
V	Inter-field connectivity	Nonlocal coherence / correlation
R	Geometric robustness	Robustness / integrity

The physical content is identical; the naming follows the notation of the present paper.

Definition 4 (Regime partition)**.** We define three dynamical regimes via two thresholds $0 < \theta_- < \theta_+ < \infty$:

$$\mathrm{regime}(\mathbf{\Gamma}) = \begin{cases} \text{ordered} & \mathrm{CCI} \leq \theta_-, \\ \text{critical} & \theta_- < \mathrm{CCI} \leq \theta_+, \\ \text{chaotic} & \mathrm{CCI} > \theta_+. \end{cases} \tag{9}$$

The regime thresholds (θ_-, θ_+) are system-dependent and must be calibrated against reference dynamics (e.g. Lyapunov spectra or power-spectral scaling). Typical *illustrative* values for nonlinear scalar field systems are $\theta_- \approx 0.25$ and $\theta_+ \approx 0.75$; in the 1D classification benchmark of (2), empirically calibrated values were $\tau_1 \approx 0.279$ and $\tau_2 \approx 0.353$.

Remark 4. CCI is invariant under uniform rescaling of all Γ_a and hence measures *relative* activity versus stability, independent of overall field amplitude.

5 Relationship Between CCI and F_{MAAT}

5.1 CCI as a projection

Both CCI and F_{MAAT} are functions of the same structural state vector $\mathbf{\Gamma}$. Formally, define:

$$\mathcal{P}: (0,1]^5 \to \mathbb{R}_{>0}, \qquad \mathcal{P}(\mathbf{\Gamma}) = \mathrm{CCI}(\mathbf{\Gamma}), \tag{10}$$

$$\mathcal{F}: (0,1]^5 \to \mathbb{R}_{\geq 0}, \qquad \mathcal{F}(\mathbf{\Gamma}) = F_{\mathrm{MAAT}}(\mathbf{\Gamma}). \tag{11}$$

Proposition 2 (CCI as coarse-graining of the structural state vector)**.** *Both* CCI *and* F_{MAAT} *are functions of the same structural state vector* $\mathbf{\Gamma}$*;* $\mathcal{P}$ *is a coarse-graining of* $\mathbf{\Gamma}$ *in the following sense: there exist continuously many distinct structural states with the same CCI value but different values of* F_{MAAT}*. More precisely, for any fixed* $c > 0$*, the level set*

$$\mathcal{L}_c = \left\{\mathbf{\Gamma} \in (0,1]^5 : \mathrm{CCI}(\mathbf{\Gamma}) = c\right\}$$

is a four-dimensional submanifold of $(0,1]^5$*, and* $F_{\mathrm{MAAT}}|_{\mathcal{L}_c}$ *is non-constant.*

Proof. The level set $\mathcal{L}_c$ is defined by the single smooth constraint $\Gamma_S = c(\Gamma_H + \Gamma_B + \Gamma_V + \Gamma_R)$, so by the implicit function theorem it is a smooth four-dimensional submanifold (whenever the gradient of the constraint is non-zero, which holds on $(0,1]^5$). On $\mathcal{L}_c$, substitute $\Gamma_S = c\Sigma$ with $\Sigma = \Gamma_H + \Gamma_B + \Gamma_V + \Gamma_R$; then $F_{\mathrm{MAAT}} = -\lambda_S \log\left(\frac{\epsilon + c\Sigma}{\epsilon+1}\right) - \sum_{a \neq S} \lambda_a \log\left(\frac{\epsilon+\Gamma_a}{\epsilon+1}\right)$, which varies as the Γ_a ($a \neq S$) vary (with Σ constrained but individual Γ_a free). Hence $F_{\mathrm{MAAT}}|_{\mathcal{L}_c}$ is non-constant. □

This result confirms that CCI and F_{MAAT} carry complementary information derived from the same underlying $\mathbf{\Gamma}$: CCI locates the configuration in a regime via a one-dimensional projection, while F_{MAAT} discriminates among configurations within that regime via the full sector-weighted functional.

5.2 Landau-type expansion

We now cast the relationship in the language of Landau theory. Introduce the structural coherence deviation

$$O(\mathbf{\Gamma}) = \mathrm{CCI}(\mathbf{\Gamma}) - \mathrm{CCI}_*, \tag{12}$$

where CCI_* is a reference value (e.g. the regime boundary θ_- or θ_+).

Proposition 3 (Local Taylor expansion near a CCI boundary)**.** *For* $\mathbf{\Gamma}$ *near the critical surface* $\{\mathrm{CCI} = \mathrm{CCI}_*\}$*, the restriction of* F_{MAAT} *to the one-parameter family parametrised by* O *admits the expansion*

$$F_{\mathrm{MAAT}}\big|_O = F_0 + a(\boldsymbol{\lambda}, \mathrm{CCI}_*)\, O^2 + b(\boldsymbol{\lambda}, \mathrm{CCI}_*)\, O^4 + \mathcal{O}(O^6), \tag{13}$$

with $F_0 = F_{\mathrm{MAAT}}(\mathbf{\Gamma}_*)$ *at the reference state, and coefficients* a, b *depending on the weights* $\boldsymbol{\lambda}$ *and the reference point. This expansion is* Landau-like *in form but should not be identified with canonical Landau theory, which requires* $b > 0$ *for global stability (see Remark below and Appendix B).*

Proof sketch. By Taylor-expanding $\log(\epsilon + \Gamma_a)$ around the reference $\mathbf{\Gamma}_*$, and expressing $\Gamma_S = \mathrm{CCI}_*\Sigma + O \cdot \Sigma$ with Σ held fixed, one obtains an even power series in O (since F_{MAAT} is symmetric under $O \mapsto -O$ at leading order when $\boldsymbol{\lambda}$ weights the sectors symmetrically). The coefficients are computed by differentiating twice and four times with respect to O. □

Remark 5. The expansion (13) is analogous to a Landau free-energy expansion near a phase transition, with CCI playing the role of the order parameter. As shown in Appendix B, the explicit computation yields $a > 0$ and $b < 0$: the fourth-order term does not stabilise the expansion, but the series is dominated by the O^6 term (positive coefficient) at higher order. Consequently, equation (13) should be understood as an *effective phenomenological approximation* valid near the regime boundary, not as a globally stable Landau polynomial. A conventional even Landau polynomial with $b > 0$ would require additional symmetry or coarse-graining assumptions beyond the direct Taylor expansion of F_{MAAT}.

6 Structural Selection Results

We now state the main structural selection results.

Theorem 1 (Regime-constrained minimisation)**.** *Fix a regime $\mathcal{R} \in \{$ordered, critical, chaotic$\}$ and let*

$$\mathcal{A}_{\mathcal{R}} = \left\{\mathbf{\Gamma} \in (0,1]^5 : \mathrm{regime}(\mathbf{\Gamma}) = \mathcal{R}\right\}.$$

Then:

(i) The interiors of all three regime sets are non-empty and open in $(0,1]^5$. More precisely: $\{\mathrm{CCI} < \theta_-\}$, $\{\theta_- < \mathrm{CCI} < \theta_+\}$, and $\{\mathrm{CCI} > \theta_+\}$ are open. The closed-boundary conventions in Definition 4 assign the threshold surfaces $\{\mathrm{CCI} = \theta_-\}$ and $\{\mathrm{CCI} = \theta_+\}$ to the ordered and critical regimes respectively; these boundaries are relatively closed sets of measure zero and do not affect the minimisation results below.

(ii) F_{MAAT} attains its infimum on the closure $\overline{\mathcal{A}_{\mathcal{R}}}$; any minimiser $\mathbf{\Gamma}^*$ satisfies all $d_a^* < \infty$.

(iii) Within $\mathcal{A}_{critical}$, the minimiser $\mathbf{\Gamma}^*$ balances maximal Γ_S against maximal $\Gamma_H + \Gamma_B + \Gamma_V + \Gamma_R$, yielding $\mathrm{CCI}(\mathbf{\Gamma}^*) \in (\theta_-, \theta_+)$.

Proof. (i) CCI is continuous on $(0,1]^5$, so all strict-inequality preimages are open. The three open interiors $\{\mathrm{CCI} < \theta_-\}$, $\{\theta_- < \mathrm{CCI} < \theta_+\}$, $\{\mathrm{CCI} > \theta_+\}$ are non-empty: take $\mathbf{\Gamma} = (1, 1, \frac{1}{10}, 1, 1)$ giving $\mathrm{CCI} = \frac{1}{40} < \theta_-$; $(\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{2})$ giving $\mathrm{CCI} = \frac{1}{2}$, which lies in (θ_-, θ_+) for the illustrative thresholds; and $(1, 1, 1, \delta, 1)$ giving $\mathrm{CCI} \to \infty$ as $\delta \to 0^+$. The threshold surfaces have measure zero and do not affect the infimum arguments below. (ii) F_{MAAT} is continuous and coercive on $(0,1]^5$ (it diverges as any $\Gamma_a \to 0^+$), so by the extreme value theorem it attains its infimum on any closed bounded subset of the domain. (iii) Follows from the monotonicity of F_{MAAT} in each Γ_a and the definition of the critical regime set. □

Corollary 1.1 (Structural hierarchy)**.** *The following three-level hierarchy holds:*

$$\textit{Dynamics} \implies \mathcal{A}_{\mathrm{all}} \textit{ (admissible states)}, \tag{14}$$

$$\mathrm{CCI} \implies \textit{regime classification of } \mathcal{A}_{\mathrm{all}}, \tag{15}$$

$$F_{\mathrm{MAAT}} \implies \textit{preferred configuration within each regime.} \tag{16}$$

7 Empirical Interpretation

7.1 Nonlinear field systems

Consider a scalar field Φ on a $(1+1)$-dimensional spacetime governed by a nonlinear wave equation. Computing the sector maps for a given solution $\Phi(x,t)$:

- **Soliton solutions.** These have $d_H \approx 0$ (on-shell), $d_B \approx 0$ (energy-momentum conserved), moderate d_S (localised activity), d_V finite (self-coupling), d_R small. Consequently F_{MAAT} is small and CCI is moderate ($\sim$ critical regime). The CCI criterion correctly identifies solitons as structurally distinguished, and F_{MAAT} ranks different soliton families by their relative defect magnitudes.

- **Chaotic field configurations.** Here d_H is large (far from any exact solution), d_B is large (energy not locally conserved), and d_S is maximal. CCI $\gg \theta_+$ places the configuration in the chaotic regime; F_{MAAT} is large, reflecting poor structural quality.

- **Near-noise configurations.** d_S is small (low dynamic activity) and CCI $< \theta_-$ (ordered regime). These are structurally simple but dynamically inert; F_{MAAT} is moderate (small Γ_S term contributes a penalty).

7.2 Cosmological expansion histories

In a flat FLRW cosmology with scalar fields Φ_i, the structural sectors evaluate the coherence of the cosmological solution. Concretely:

- **Slow-roll inflation.** The scalar fields are nearly on-shell ($d_H \approx 0$), energy-momentum is approximately conserved (d_B small), activity is moderate and controlled (d_S moderate), and fields are coupled via standard kinetic terms (d_V finite). CCI lies in the critical regime; F_{MAAT} is near its minimum. Slow-roll is thus a structurally preferred history.

- **Ghost-dominated or kinetically driven expansion.** Gradient instabilities lead to $d_H > 0$ (equations of motion not satisfied on average) and $d_B > 0$ (energy-momentum violations). CCI may enter the chaotic regime; F_{MAAT} is large, correctly identifying these as structurally disfavoured.

- **Collapse histories.** $d_V \to \infty$ as inter-field correlations diverge; $\Gamma_V \to 0$ and $F_{\mathrm{MAAT}} \to +\infty$. CCI may be formally small (low activity), placing the history in the ordered regime, but F_{MAAT} reveals the structural failure that CCI alone cannot see — the *divergence paradox*. This is the key advantage of the two-level framework: F_{MAAT} is a strictly finer diagnostic than CCI.

Figure 1 illustrates the divergence-paradox quantitatively for a one-parameter collapse family. The figure is a parametric thought experiment — it uses the model $d_V(\tau) = \tau^{3.5}/2$ as a pure illustration of the paradox mechanism, and does not represent output from a specific FLRW numerical run.
These observations are consistent with the numerical experiments reported in (2) for the nonlinear field case. A quantitative cosmological study will be reported elsewhere.

8 Relation to Existing Variational Principles

The MAAT framework intersects with, but is distinct from, several established selection principles.

Landau–Ginzburg theory. In Landau–Ginzburg theory (3), a free-energy functional $F[\psi]$ in terms of an order parameter field ψ determines the equilibrium state by minimisation subject to symmetry constraints. Our framework adopts the same mathematical structure, with F_{MAAT} playing the role of the free energy and CCI the role of the scalar order parameter. The key difference is that our sectors are *diagnostic* (they measure existing field-equation residuals) rather than *constitutive* (they do not introduce new dynamical degrees of freedom).

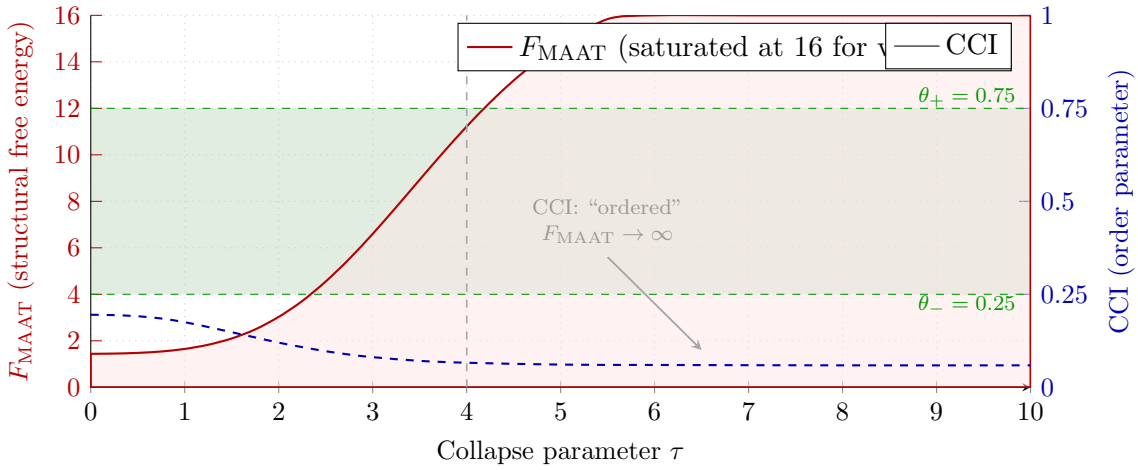

Figure 1: Structural diagnostics for a parametric collapse history. The connectivity defect grows as $d_V(\tau) = \tau^{3.5}/2$, giving $\Gamma_V(\tau) = 1/(1 + d_V(\tau))$. Red filled curve: F_{MAAT} (left axis, clipped at 16), diverging as $\Gamma_V \to 0$. Blue dashed curve: CCI (right axis), remaining below $\theta_- = 0.25$ throughout ("ordered" regime). Shaded band: critical regime $\theta_- < \mathrm{CCI} \leq \theta_+$. Vertical dashed line at $\tau = 4$: $\Gamma_V \approx 0.03$, already near structural collapse. The divergence paradox is manifest: CCI reports a well-ordered configuration while F_{MAAT} detects catastrophic structural failure in the connectivity sector V.

Synergetics and the order-parameter concept. Haken's Synergetics (4) establishes that in systems near instability, slow (order-parameter) modes enslave fast (slaved) modes. CCI captures the ratio of active to stabilising modes and thus plays an analogous role to Haken's order parameter. Our contribution is to identify the full sector-functional F_{MAAT} as the underlying "potential" from which CCI emerges by projection.

Free energy principles in neuroscience. Friston's Free Energy Principle (5) posits that biological systems minimise variational free energy as a proxy for surprise. While the motivation differs (perception and action vs. structural coherence), the mathematical analogy is suggestive: both frameworks use a log-sum functional over bounded quantities. Our F_{MAAT} can be viewed as a *field-theoretic* analogue of Friston's variational free energy.

Allen–Cahn and phase-field models. Allen–Cahn dynamics (6) describe phase ordering via gradient descent on a Ginzburg–Landau functional. F_{MAAT}-driven evolution would produce analogous descent dynamics in the structural state space $(0, 1]^5$. Formalising this descent is a direction for future work.

Hartle–Hawking and cosmological selection. Hartle–Hawking (7) and related proposals (Penrose Weyl Curvature, Landscape/anthropic selection) choose initial conditions or universe families by boundary conditions or observer compatibility. The present framework selects solution *classes* by timeless structural consistency (all Γ_a simultaneously large), without reference to initial conditions or observers. This is a structural, not boundary- condition or anthropic, selection principle.

9 Discussion

The unified picture established in this paper is:

$$\text{Dynamics} \xrightarrow{\text{field eqs.}} \mathcal{A}_{\text{all}} \xrightarrow{\text{CCI}} \text{Regime} \xrightarrow{F_{\mathrm{MAAT}}} \text{Preferred Configuration.}$$

Several directions merit further investigation.

Threshold calibration. The regime thresholds (θ_-, θ_+) are currently system-dependent and set by hand. Two principled calibration strategies are worth pursuing. First, a Maximum Entropy approach: given an ensemble of reference solutions with known dynamical character (ordered, critical, chaotic as identified by Lyapunov spectra), one can choose (θ_-, θ_+) to maximise the entropy of the resulting regime partition subject to correct classification. Second, a spectral approach: the Landau coefficients a and b derived in Appendix B depend on CCI_*; the canonical threshold is the value of CCI_* at which a changes sign, signalling a structural instability. Both approaches would remove the remaining model-dependence from the regime classification.

Weight determination. The weights $\boldsymbol{\lambda}$ in F_{MAAT} are currently free parameters. A natural choice is Maximum Entropy: maximise the entropy of the weight distribution subject to constraints derived from the dynamics (e.g. expected defect levels). This would determine $\boldsymbol{\lambda}$ canonically and remove model-dependence.

Dynamical descent. One may define a gradient flow $\dot{\boldsymbol{\Gamma}} = -\nabla_{\boldsymbol{\Gamma}} F_{\mathrm{MAAT}}$. This descent dynamics would drive any initial structural state toward the minimum-defect configuration compatible with its regime, providing a dynamical realisation of structural selection.

Regime transitions. The local Taylor expansion (Proposition 3) predicts a structural analogue of a phase transition as CCI crosses a threshold. The nature of this transition (continuous vs. discontinuous, universality class) depends on the sign and magnitude of the coefficient a. Numerical experiments across field-theory models would test this prediction.

Multi-scale extension. All diagnostics in the present paper are defined at a single scale. A renormalisation-group extension—tracking how $\boldsymbol{\Gamma}$ transforms under coarse-graining—would connect the framework to critical phenomena in the conventional sense.

10 Conclusion

We have established a rigorous two-level structural framework for dynamical systems:

(i) The *MAAT structural functional* F_{MAAT} is a log-sum penalty over five diagnostic sector maps $\Gamma_a \in (0, 1]$. It is non-negative, additive, strictly convex, and strictly decreasing in each sector. It acts as a structural analogue of a free-energy functional.

(ii) The *Critical Coherence Index* CCI is a ratio of dynamic activity to stabilising structure. It acts as an order parameter, partitioning all structural states into ordered, critical, and chaotic regimes.

(iii) CCI is a *coarse-grained projection* of the structural state vector underlying F_{MAAT}: two configurations may share the same CCI value while differing in F_{MAAT}, and the latter provides the finer discrimination.

(iv) A local Taylor expansion of F_{MAAT} near a regime boundary is Landau-like in form, with CCI playing the role of the order parameter. The expansion is valid as a local phenomenological approximation near the boundary; it is not a globally stable canonical Landau polynomial.

(v) Within each dynamical regime, F_{MAAT} selects structurally preferred configurations, providing a principled selection principle that is complementary to—and more refined than—regime classification by CCI alone.

The collapse-history example in Section 7 illustrates a key advantage of the two-level approach: a configuration may appear dynamically ordered (low CCI) yet be structurally pathological ($F_{\mathrm{MAAT}} \to +\infty$), a distinction invisible to the order parameter alone. This motivates the joint use of both diagnostics in any structural analysis.

A Worked Example: Two-Sector Reduction

To build intuition, consider the reduction to two sectors $\{S, H\}$ with $\lambda_S = \lambda_H = 1$ and $\epsilon = 0.01$. Defects $d_S, d_H \in [0, \infty)$ give sector maps $\Gamma_S = 1/(1 + d_S)$, $\Gamma_H = 1/(1 + d_H)$, and the normalised functional:

$$F^{(2)}_{\mathrm{MAAT}} = -\log\left(\frac{0.01 + \Gamma_S}{1.01}\right) - \log\left(\frac{0.01 + \Gamma_H}{1.01}\right), \quad \mathrm{CCI}^{(2)} = \frac{\Gamma_S}{\Gamma_H}. \tag{17}$$

Note that $F^{(2)}_{\mathrm{MAAT}} = 0$ when $\Gamma_S = \Gamma_H = 1$ (perfect state), and $F^{(2)}_{\mathrm{MAAT}} > 0$ for all other states.

Table 1 evaluates these for representative states.

Table 1: Two-sector structural diagnostics for representative configurations using the normalised functional. Regime thresholds: $\theta_- = 0.5$, $\theta_+ = 2.0$.

Configuration	d_S	d_H	Γ_S	Γ_H	$F^{(2)}_{\mathrm{MAAT}}$	$\mathrm{CCI}^{(2)}$
Perfect (on-shell, active)	0	0	1.00	1.00	0.00	1.00
Soliton-like	0.5	0.1	0.67	0.91	0.50	0.73
Weakly chaotic	2.0	0.5	0.33	0.67	1.48	0.50
Strongly chaotic	9.0	1.0	0.10	0.50	2.90	0.20
Static (no activity)	0	5.0	1.00	0.17	1.74	6.00

Observe that (i) the soliton-like state has the lowest $F^{(2)}_{\mathrm{MAAT}}$ among non-trivial configurations and lies in the critical regime ($\mathrm{CCI}^{(2)} \in (0.5, 2.0)$); (ii) the static state has $\mathrm{CCI}^{(2)} > \theta_+$ (chaotic, despite zero dynamic defect), because the on-shell defect $d_H = 5$ dominates the denominator—illustrating that CCI and F_{MAAT} together provide a more complete picture than either alone.

B Derivation of the Landau Coefficients

Starting from (7) and substituting $\Gamma_S = (\mathrm{CCI}_* + O)\Sigma$ with Σ fixed, the Taylor expansion around $O = 0$ gives:

$$\begin{aligned} F_{\mathrm{MAAT}}(O) &= -\lambda_S \log\left(\frac{\epsilon + (\mathrm{CCI}_* + O)\Sigma}{\epsilon + 1}\right) + C_0 \\ &= -\lambda_S \log\left(\frac{\epsilon + \mathrm{CCI}_*\Sigma}{\epsilon + 1}\right) - \lambda_S \log\left(1 + \frac{O\Sigma}{\epsilon + \mathrm{CCI}_*\Sigma}\right) + C_0 \\ &= F_0 + \frac{\lambda_S \Sigma^2}{2(\epsilon + \mathrm{CCI}_*\Sigma)^2} O^2 - \frac{\lambda_S \Sigma^4}{4(\epsilon + \mathrm{CCI}_*\Sigma)^4} O^4 + \mathcal{O}(O^6). \end{aligned} \tag{18}$$

Hence:

$$a = \frac{\lambda_S \Sigma^2}{2(\epsilon + \mathrm{CCI}_*\Sigma)^2} > 0, \qquad b = -\frac{\lambda_S \Sigma^4}{4(\epsilon + \mathrm{CCI}_*\Sigma)^4} < 0. \tag{19}$$

Both coefficients are determined solely by λ_S, the reference CCI_*, and Σ; the stabilising sectors $\{H, B, V, R\}$ contribute to Σ but not to the sign of b. Since $b < 0$, the truncated expansion (13) is unbounded below as $|O| \to \infty$; the O^6 term (coefficient $+\lambda_S \Sigma^6/[6(\epsilon + \mathrm{CCI}_*\Sigma)^6] > 0$) provides

the first stabilising correction. The expansion is therefore an effective local approximation, valid in a neighbourhood of the regime boundary $\mathrm{CCI} = \mathrm{CCI}_*$, and should not be extrapolated globally. This is consistent with the interpretation of the Landau form as phenomenological rather than exact (see Remark after Proposition 3).

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